Transfer: waves and signals

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Learning target: Choose appropriate wave models and explain what a signal establishes.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 15.*
2500 m; range is half the full path.

02. Worked example: explain the model
Round-trip timing plus propagation speed gives range; a separate frequency comparison helps determine radial motion.

03. Read and interpret the model
Amplitude: size from equilibrium; wavelength: distance between matching phase points.

04. Apply the physics
10 m/s; range 20 m.

05. Explain your reasoning
No; amplitude/intensity change is distinct from frequency.

06. Review: connect an earlier idea *Review: Unit 6, Day 6.*
The 6 cm wave generally spreads more because wavelength is larger relative to the same opening.

07. Exit check
Toward normal; frequency unchanged at stationary boundary.

08. STRETCH: test the model
Sound reflector 85 m; light reflector 300 m. Shorter time need not mean shorter range when signal speeds differ.

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01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 15.*
At the same wave speed, equal range gives equal out-and-back time. Doppler shift depends on radial velocity, so motion can differ.

02. Worked example: explain the model
Round-trip timing plus propagation speed gives range; a separate frequency comparison helps determine radial motion.

03. Read and interpret the model
Amplitude: size from equilibrium; wavelength: distance between matching phase points.

04. Apply the physics
1.8×10^{6} m;λ=0.10 m.

05. Explain your reasoning
No; amplitude/intensity change is distinct from frequency.

06. Review: connect an earlier idea *Review: Unit 6, Day 6.*
Wavelength becomes 1/4; typically less spreading when the opening is large relative to the new wavelength.

07. Exit check
Thermal peak halves by Wien; sound λhalves at fixed sound speed.

08. STRETCH: test the model
1500 and 1440 m; radial velocity -300 m/s. Electromagnetic propagation remains about 3×10^{8} m/s in air under the model.